Band-1 Kummer–Vandiver: Explicit-p Verification Probe (Collapse List 2.22)

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Band-1 Kummer–Vandiver: Explicit Verification Bound p (Probe Only)

Honesty stamp. This note records the **published** exhaustive verification bound for **irregular primes** and the **Kummer–Vandiver conjecture**, and certifies a **tiny on-box Bernoulli irregularity re-check** for $p \leq 200$. It does **not** prove the Vandiver conjecture. It does **not** claim a Clay-millennial or ToE result. **GATE-04** is locked out of scope.

> **Kummer–Vandiver remains OPEN.** Finite-range verification $\neq$ conjecture. **Do not claim Vandiver proved.**

Abstract

Published peer-reviewed bound: for every odd prime $p < 2^{31} = 2\,147\,483\,648$, Hart–Harvey–Ong computed the irregular indices, verified the Kummer–Vandiver conjecture, and checked $\lambda_p = \nu_p = i_p$ (*Mathematics of Computation* **86** (2017), 3031–3049; DOI 10.1090/mcom/3211; arXiv:1605.02398). **Zero** counterexamples in this range.

Box method demo / re-check: Bernoulli irregularity scan for all odd primes $p \leq 200$ (**45** primes; **8** irregular; **0** mismatches vs classical table; wall ≈ 0.05 s). This window is **not** a new world record (orders of magnitude below 2^{31}). Full $V_{p,r}$ Vandiver checks were **not** re-run on box at scale.

Problem (Band-1 collapse item 2.22 / H-2.22 / NT-VANDIVER)

1. Cite the current **published** explicit verification bound for irregular primes / Kummer–Vandiver without invention.
2. Re-run a classical Bernoulli congruence scan on a **certified tiny box window**.
3. Keep GATE-04, Clay rhetoric, and "Vandiver proved" out of the claim surface.

Statement (literature)

Let p be an odd prime. An even index $r \in \{2, 4, \ldots, p-3\}$ is **irregular** for p if $B_r \equiv 0 \pmod{p}$ (B_r the r-th Bernoulli number). The **index of irregularity** i_p is the number of such r; p is **regular** if $i_p = 0$ and **irregular** if $i_p > 0$.

The **Kummer–Vandiver conjecture** asserts that p does **not** divide the class number of the maximal real subfield of $\mathbf{Q}(\zeta_p)$.

Literature ledger (2024–2026 web check; not re-proved on box)

Year	Bound	Authors	Status
1850	$p \leq 43$	Kummer	historical
1937	613	Vandiver	historical
2001	12 000 000	Buhler–Crandall–Ernvall–Metsänkylä–Shokrollahi	published
2011	163 577 856	Buhler & Harvey, *Math. Comp.* 80	published
2017	$2^{31} = 2\,147\,483\,648$	**Hart, Harvey & Ong**, *Math. Comp.* 86	**published record**
2024	n/a (distribution)	Qin, *Acta Math. Sinica* Chinese Ser. 67	adjacent theory — **not** an exhaustive p-bound
2024–2026	claims past 2^{31}	web sweep	**none elevated** above Hart–Harvey–Ong

Canonical citation for $p_{\text{pub}} < 2^{31}$. W. Hart, D. Harvey, W. Ong, *Irregular primes to two billion*, Mathematics of Computation **86** (2017), 3031–3049. arXiv:1605.02398. DOI 10.1090/mcom/3211. Also verified: Kummer–Vandiver for each such p, and $\lambda_p = \nu_p = i_p$. Prior published bound: Buhler & Harvey, *Math. Comp.* **80** (2011), DOI 10.1090/S0025-5718-2011-02461-0 (to 163 577 856).

Hygiene. Qin (2024) proposes distribution heuristics related to Vandiver / irregular primes; it is **not** treated as extending the exhaustive computational bound. No peer-reviewed exhaustive irregular/Vandiver verification beyond Hart–Harvey–Ong was elevated in the Sep 2026 sweep. Wikipedia (accessed Sep 2026) agrees: verified for $p < 2^{31}$.

Method (box)

1. **Bernoulli.** Recurrence with $B_1 = -1/2$ up to B_{200}; sanity $B_2 = 1/6$, $B_{12} = -691/2730$.
2. **Irregular indices.** For each odd prime $p \leq 200$, list even $r \in \{2, \ldots, p-3\}$ with numerator of B_r divisible by p and denominator not.
3. **Unit tests.** First irregular = 37 with $r = 32$; set of irregular primes ≤ 200 matches classical table $\{37, 59, 67, 101, 103, 131, 149, 157\}$.
4. **Label.** Method demo / re-check — **not** a record attempt; **not** a full $V_{p,r}$ Vandiver re-verification.

Results

Published p (cited)

Quantity	Value
p_{pub}	**all odd primes $p < 2^{31}$**
Numeric	2 147 483 648
Source	Hart–Harvey–Ong, *Math. Comp.* 86 (2017)
Also checked	Kummer–Vandiver; $\lambda_p = \nu_p = i_p$
Counterexamples	**0**
Box beats published?	**No**

Box window (certified tonight)

Quantity	Value
Box $p_{\max}$	200
Odd primes checked	**45**
Irregular	**8**
Regular	**37**

Quantity	Value
Failures vs classical table	**0**
First irregular	37, $r = 32$
Wall (box)	≈ 0.05 s
Full $V_{p,r}$ on box	**not run**

Sample irregular pairs (box)

p	i_p	irregular indices r
37	1	32
59	1	44
67	1	58
101	1	68
103	1	24
131	1	22
149	1	130
157	2	62, 110

Why Band-1 vs conjecture

- **Band-1 (this probe):** cite explicit published p-bound; certify a finite Bernoulli re-check window; method hygiene.
- **Conjecture (OPEN):** Kummer–Vandiver for **all** primes — **not proved here**. Washington heuristics suggest counterexamples (if any) grow like $\log\log x$; absence below 2^{31} is consistent but **not** a proof.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/kummer_vandiver/`
- Run: `python3 kummer_vandiver_demo.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `METHOD.md`

Conclusion

Published verified range: all odd primes $p < 2^{31} = 2\,147\,483\,648$ (Hart–Harvey–Ong 2017) — irregular indices + Kummer–Vandiver + $\lambda_p = \nu_p = i_p$; **0** counterexamples. **Box window:** Bernoulli irregularity for $p \leq 200$ matches classical table (8 irregular). **Kummer–Vandiver: OPEN**. No Clay / ToE claim. GATE-04 locked. **Do not claim Vandiver proved.**